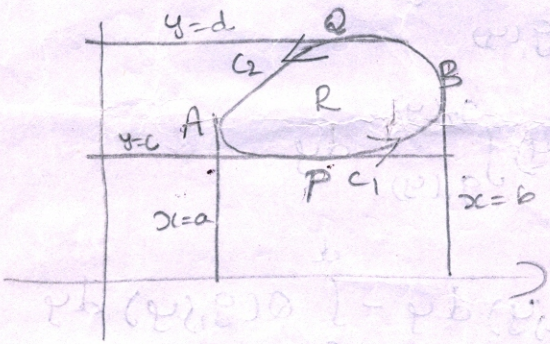
Green's theorem in the plane

Statement: If $P(x,y)$ and $Q(x,y)$ be two continuous functions having continuous partial derivatives in a region R of the xy-plane, bounded by a simple closed curve C, then

$$\oint_C (P\,dx + Q\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

Proof: we shall consider the region R of the xy-plane bounded by a single closed curve C, such that any line drawn parallel to either coordinate axis meets the boundary C in at most two points



Proof of the theorem is not included in the syllabus

Let us assume that the curve C is made up of two curves APB and BQA, denoted by C_1 and C_2.

Let the eq^{n}s be $y = f_1(x)$ and $y = f_2(x)$ respectively

Now $$\iint_R \frac{\partial P}{\partial y}\,dx\,dy = \int_{x=a}^{x=b} \int_{y=f_1(x)}^{y=f_2(x)} \frac{\partial P}{\partial y}\,dy\,dx$$

$$= \int_a^b [P(x,y)]_{f_1(x)}^{f_2(x)}\,dx$$

$$= \int_a^b [P(x,f_2) - P(x,f_1)]\,dx$$

$$= -\left[\int_a^b P(x,f_1)\,dx - \int_a^b P(x,f_2)\,dx\right]$$

$$= -\left[\int_a^b P(x,f_1)\,dx + \int_b^a P(x,f_2)\,dx\right] = -\left(\int_{C_1} P(x,y)\,dx + \int_{C_2} P(x,y)\,dx\right)$$

$$= -\oint_C P(x,y)\,dx$$

$$= -\oint_C P\,dx \qquad \rightarrow (1)$$

Now let us assume that the curve is made up of two curves QAP and PBQ denoted by C_1 and C_2 and its equations are

$x = g_1(y)$ and $x = g_2(y)$ respectively

Thus $$\iint_R \frac{\partial Q}{\partial x}\,dx\,dy = \int_{y=c}^{y=d}\int_{x=g_1(y)}^{x=g_2(y)} \frac{\partial Q}{\partial x}\,dx\,dy$$

$$= \int_c^d Q(x,y)\Big]_{g_1(y)}^{g_2(y)}\,dy$$

$$= \int_c^d Q(g_2,y)\,dy - \int_c^d Q(g_1,y)\,dy$$

$$= \int_c^d Q(g_2,y) + \int_d^c Q(g_1,y)\,dy$$

$$= \int_{C_1} Q(x,y)\,dx + \int_{C_2} Q(x,y)\,dy$$

$$= \oint_C Q(x,y)\,dy$$

$$= \oint_C Q\,dy \qquad \rightarrow (2)$$

Proof of the theorem is not included in the syllabus

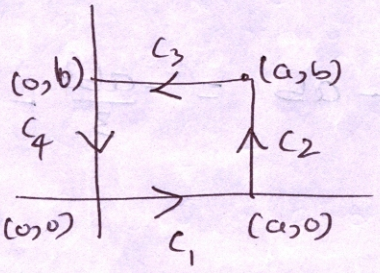
From (1) and (2)

$$\oint_C (P\,dx + Q\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)dx\,dy$$

Note:- Green's theorem gives us a relation b/w the ~~plane~~ double integral and the line integral.

① Verify Green's theorem in the plane for $\int_C (x^2+y)dx - xy^2 dy$ taken around the boundary of the rectangle whose vertices are $(0,0)$ $(a,0)$ (a,b) and $(0,b)$

Soln:



By Green's theorem, we have

$$\oint_C (P dx + Q dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

Here $P = x^2+y \qquad Q = -xy^2$

$$\oint_C (P dx + Q dy) = \int_{C_1} (P dx + Q dy) + \int_{C_2} (P dx + Q dy) + \int_{C_3} (P dx + Q dy) + \int_{C_4} (P dx + Q dy).$$

Along C_1 : $y=0$, $dy=0$, x : 0 to a

$$\therefore \int_{C_1} (x^2+y)dx - x^2y\,dy = \int_{x=0}^{x=a} x^2 dx = \frac{a^3}{3}$$

Along C_2 : $x=a$, $dx=0$: y : 0 to b

$$\int_{C_2} (x^2+y)dx - x^2y\,dy = \int_{y=0}^{y=b} -a^2y^2\,dy = \frac{-ab^3}{3}$$

Along C_3 : $y=b$ $\quad x$: a to 0

$dy=0$

$$\int_{C_3}(x^2+y)\,dx - xy^2\,dy = \int_a^0 (x^2+b)\,dx = -\frac{a^3}{3} - ab$$

Along C_4 $\quad x=0$: y: b to 0

$dx=0$

$$\int_{C_4}(x^2+y)\,dx - xy^2\,dy = \int_b^0 0\,dy = 0$$

$$\therefore \int_C P\,dx + Q\,dy = \frac{a^3}{3} - \frac{ab^3}{3} - \frac{a^3}{3} - ab = -\frac{ab^3}{3} - ab$$

Also $\quad P = x^2+y \qquad Q = -xy^2$

$$\frac{\partial P}{\partial y} = 1 \qquad \frac{\partial Q}{\partial x} = -y^2$$

$$\iint_R \left(\frac{\partial P}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \int_{x=0}^{x=a}\int_{y=0}^{y=b} (-y^2-1)\,dy\,dx$$

$$= \int_0^b (-y^2-1)\,dy \times \int_0^a dx$$

$$= -\frac{ab^3}{3} - ab$$

From ① and ② Green's theorem is verified.

② Verify Green's theorem in the plane for

$\oint_C [(3x^2-8y^2)\,dx + (4y-6xy)\,dy]$ where C is the region bounded by the parabolas $y^2=x$ and $y=x^2$

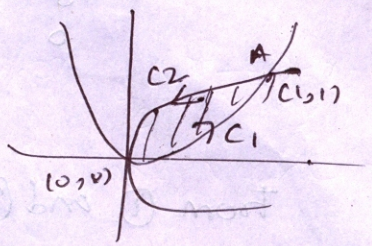
Soln: Here $P = 3x^2 - 8y^2$, $Q = 4y - 6xy$

The curves $y^2 = x$ and $y = x^2$ intersect each other at $(1,1)$

C is bounded by the curves

C_1: $y = x^2$ from $(0,0)$ to $(1,1)$

C_2: $y^2 = x$ from $(1,1)$ to $(0,0)$



Along $C_1 \Rightarrow$ $y = x^2$ and $x \to 0$ to 1

$dy = 2x\,dx$

$$\therefore \int_{C_1} (P\,dx + Q\,dy) = \int_0^1 (3x^2 - 8x^4)\,dx + (4x^2 - 6x^3)2x\,dx$$

$$= \int_0^1 (3x^2 + 8x^3 - 20x^4)\,dx$$

$$= \left[x^3 + 2x^4 - 4x^5\right] = -1$$

Along $C_2 \Rightarrow x = y^2 \Rightarrow dx = 2y\,dy$ and y varies from 1 to 0

$$\int_{y=1}^{y=0} (3y^4 - 8y^2)2y\,dy + (4y - 6y^3)\,dy = \int_1^0 (6y^5 - 22y^3 + 4y)\,dy$$

$$= \left[y^6 - \tfrac{11}{2}y^4 + 2y^2\right]_1^0$$

$$= -\left[1 - 11/2 + 2\right] = 5/2$$

$$\therefore \oint_C (P\,dx + Q\,dy) = -1 + \frac{5}{2} = 3/2 \quad \to (1)$$

In the Region R, x varies from $x = 0$ to $x = 1$, y varies from $y = x^2$ to $y = \sqrt{x}$

$$\therefore \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \int_{x=0}^{x=1} \int_{y=x^2}^{y=\sqrt{x}} (-6y + 16y)\,dy\,dx$$

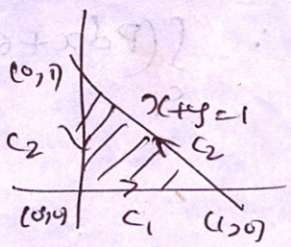
$$= \int_0^1 \int_{x^2}^{x} \log y \, dy \, dx$$

$$= \int_0^1 5y^2 \Big]_{x^2}^{x} dx = \int_0^1 (5x - 5x^4) dx$$

$$= \left[\frac{5x^2}{2} - x^5\right]_0^1 = \frac{5}{2} - 1 = 3/2 \quad \rightarrow (2)$$

from (1) and (2), C.o.T is verified.

(3) Using Green's theorem in the plane, evaluate $\int (2x^2 - y^2)\,dx + (x^2 + y^2)\,dy$ where C is the region bounded by $x = 0$, $y = 0$, $x + y = 1$



Soln. By C.o.T we have

$$\int_C (P\,dx + Q\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

$$\Rightarrow \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \iint_R (2x + 2y)\,dx\,dy$$

$$= \int_{x=0}^{x=1} \int_{y=0}^{1-x} 2(x+y)\,dy\,dx$$

$$= \int_0^1 2xy + y^2 \Big]_0^{1-x} dx$$

$$= \int_0^1 \left[2x(1-x) + (1-x)^2\right] dx$$

$$= \int_0^1 (1 - x^2)\,dx = \left[x - \frac{x^3}{3}\right] = 1 - \frac{1}{3} = 2/3$$

(4) Show that the area bounded by a simple closed curve C is given by $\frac{1}{2}\oint_C x\,dy - y\,dx$. Hence show that the area of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is πab.

Soln.. WKT :: $\oint_C P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$

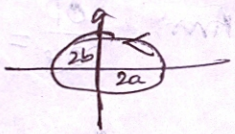
$P = -y \qquad Q = x$

$\therefore \frac{1}{2}\oint x\,dy - y\,dx = \frac{1}{2}\iint_R (1+1)\,dx\,dy$

$= \iint_R dx\,dy = \text{Area}.$

To find the area of the ellipse, the parametric eqn of ellipse are taken i.e $x = a\cos\theta \qquad y = b\sin\theta$

$dx = -a\sin\theta\,d\theta \qquad dy = b\cos\theta\,d\theta$



$$\text{Area} = \frac{1}{2}\oint_C \{x\,dy - y\,dx\}$$

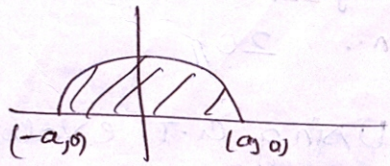
$$= \frac{1}{2}\oint_C \{a\cos\theta\, b\cos\theta - b\sin\theta(-a\sin\theta)\}\,d\theta$$

$$= \frac{1}{2}ab\oint_0^{2\pi} (\cos^2\theta + \sin^2\theta)\,d\theta = \frac{1}{2}ab\cdot 2\pi$$

$$= \pi ab \;//$$

(5) Apply Green's theorem to evaluate $\int_C (2x^2 - y^2)\,dx + (x^2+y^2)\,dy$ where C is the boundary of the area enclosed by the x-axis and upper half of the circle $x^2+y^2=a^2$

$$\iint \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \int_{x=-a}^{x=a}\int_{y=0}^{\sqrt{a^2-x^2}} (2x+2y)\,dy = \frac{4a^3}{3}$$



$$= 2\int_{-a}^{a}\left(xy+\frac{y^2}{2}\right)_0^{\sqrt{a^2-x^2}}dx$$

$$= 2\int_{-a}^{a} x\sqrt{a^2-x^2}\,dx + \int_{-a}^{a}(a^2-x^2)\,dx \quad \left[\because \int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx \text{ if } f \text{ is even}, \; = 0 \text{ if } f \text{ is odd}\right]$$

$$= 0 + 2\int_0^a (a^2-x^2)\,dx = 4\frac{a^3}{3}$$

(1) Using Green's theorem evaluate

$\int_C e^{-x}(\sin y\,dx + \cos y\,dy)$ where C is the rectangle with vertices $(0,0), (\pi,0), (\pi, \pi/2)$ and $(0,\pi/2)$

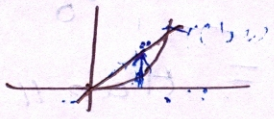
Ans: $\frac{\partial P}{\partial y} = e^{-x}\cos y \qquad \frac{\partial Q}{\partial x} = -e^{-x}\cos y$

$x \to 0$ to π, $y \to 0$ to $\pi/2$

$= 2[e^{-\pi} - 1]$

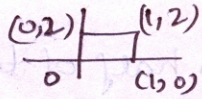
(2) Verify G.T in the plane for $\oint_C (xy+y^2)\,dx + x^2\,dy$ where C is the closed curve bounded by $y=x$ and $y=x^2$

Soln:- $\frac{-1}{20}$



(3) Verify G.T for $\oint_C (3x^2-8y^2)\,dx + 2y(2-3x)\,dy$ where C is the rectangular area ~~bound~~ enclosed by the lines $x=0, x=1, y=0$ $y=2$.

Soln:- 20 ||



(4) Using G.T evaluate $\int_C (y-\sin x)\,dx + \cos x\,dy$ where C is the Plane triangle enclosed by the lines $y=0$, $x=\pi/2$, $y=\frac{2}{\pi}x$

Soln:- $-\left(\frac{\pi}{4}+\frac{2}{\pi}\right)$

